

Fundamental Equations

Schrodinger Equation	$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi$
Time-independent Schrodinger Equation	$\hat{H} \psi = E \psi, \quad \Psi(\vec{x}, t) = \psi(\vec{x}) e^{-iEt/\hbar}$
Hamiltonian operator	$\hat{H} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m} \nabla^2 + V$
Momentum operator	$\hat{p} = -i\hbar \nabla, \quad \text{1D: } \hat{p} = -i\hbar \frac{\partial}{\partial x}$
Time dependence expectation value	$\frac{d\langle \hat{Q} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{Q}] \rangle + \left\langle \frac{\partial \hat{Q}}{\partial t} \right\rangle$
Generalized uncertainty principle	$\sigma_A \sigma_B \geq \left \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right $
Heisenberg uncertainty principle	$\sigma_x \sigma_p \geq \frac{\hbar}{2}$
Canonical commutator	$[\hat{x}, \hat{p}] = i\hbar$
Angular momentum	$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z, \quad [\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x, \quad [\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$
Pauli matrices	$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Minimal coupling	$i\hbar \frac{\partial \Psi}{\partial t} = \left[\frac{1}{2m} (\vec{p} - q\vec{A})^2 + qV \right] \Psi$
Pauli equation	$i\hbar \frac{\partial}{\partial t} \chi = \left[\frac{1}{2m_e} (\vec{p} - e\vec{A})^2 \mathbb{1} + eV \mathbb{1} + \frac{g\mu_B}{2} \hat{\vec{\sigma}} \cdot \vec{B} \right] \chi$ with spinor $\chi = \begin{pmatrix} \varphi_{\uparrow} \\ \varphi_{\downarrow} \end{pmatrix}$

Operators & Dirac notation

Bra, ket in $\mathbb{R}^n$	$\langle \alpha = \alpha \rangle^\dagger = (a_1^*, \dots, a_n^*), \quad \alpha \rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$
Inner product in $\mathbb{R}^n$ and L^2	$\langle \alpha \beta \rangle = \sum_j a_j^* b_j, \quad \langle f g \rangle = \int_a^b f^* g \, dx$
Schwartz inequality	$ \langle \alpha \beta \rangle ^2 \leq \langle \alpha \alpha \rangle \langle \beta \beta \rangle$
Triangle inequality	$\ \alpha \rangle + \beta \rangle \ \leq \ \alpha \rangle \ + \ \beta \rangle \ $
Eigenstate expansion	$ \psi \rangle = \sum_n c_n \psi_n \rangle \quad \text{with } c_n = \langle \psi_n \psi \rangle$
Projection operator	$\hat{P}_\alpha = \alpha \rangle \langle \alpha $
Commutator	$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$
Anti-commutator	$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$

Fundamental Constants

Mass of an electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Mass of a proton	$m_p \approx 1836 \, m_e$
Charge of an electron	$-e = -1.602 \times 10^{-19} \text{ C}$
Electronvolt	$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
Atomic mass unit	$1 \text{ Dalton} = 1.66 \times 10^{-27} \text{ kg} \approx \frac{5}{3} \times 10^{-27} \text{ kg}$
Bohr magneton	$\mu_B = 9.3 \times 10^{-24} \text{ J/T} = 5.8 \times 10^{-5} \text{ eV/T}$
Joule in SI units	$[J = \text{kg m}^2/\text{s}^2]$
Boltzmann constant	$k_B = 1.38 \times 10^{-23} \text{ J/K} = 8.62 \times 10^{-5} \text{ eV/K}$
Planck's constant	$h = 6.63 \times 10^{-34} \text{ J s} = 4.14 \times 10^{-15} \text{ eV s}$
Reduced Planck's constant	$\hbar = \frac{h}{2\pi} = 1.055 \times 10^{-34} \text{ J s} = 6.582 \times 10^{-16} \text{ eV s}$
Speed of light	$c = 3 \times 10^8 \text{ m/s}$
Conversion $E = hf = \frac{hc}{\lambda}$	$hc = 1240 \text{ eV nm}$
Rest energy electron	$m_e c^2 = 0.511 \text{ MeV}$

Bound States Common Potentials

1D Infinite well

$$V(x) = 0 \text{ if } 0 < x < L \\ V(x) = \infty \text{ otherwise}$$

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right),$$

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}, \quad n = 1, 2, 3, \dots$$

3D Infinite well

$$V = 0 \text{ if } 0 < x_i < L_i, \\ V = \infty \text{ otherwise}$$

$$\psi_{n_x, n_y, n_z} = \sqrt{\frac{8}{L_x L_y L_z}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right)$$

$$E_n = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right), \quad n_x(n_y, n_z) = 1, 2, 3, \dots$$

1D δ -function

$$\text{potential well} \\ V(x) = -\alpha \delta(x)$$

$$\psi(x) = \sqrt{\kappa} e^{-\kappa|x|}, \quad \kappa = \frac{m\alpha}{\hbar^2}, \quad E = -\frac{m\alpha^2}{2\hbar^2}$$

1D Linear potential

$$V(x) = e\tilde{E}x \text{ if } 0 < x < L \\ V(x) = \infty \text{ otherwise}$$

$$\psi_n(x) = C \text{Ai}(z) + D \text{Bi}(z), \quad \frac{D}{C} = -\frac{\text{Bi}(z_0)}{\text{Ai}(z_0)}$$

$$\varepsilon = \frac{E}{E_1^\infty}, \quad \nu_L = \frac{e\tilde{E}L}{E_1^\infty}, \quad z = \left(\frac{\pi}{\nu_L}\right)^{\frac{2}{3}} \left(\frac{x}{L}\nu_L - \varepsilon\right)$$

$$E_n \longleftarrow \text{Ai}(z_0) \text{Bi}(z_L) - \text{Ai}(z_L) \text{Bi}(z_0) = 0$$

$$z_0 = -\left(\frac{\pi}{\nu_L}\right)^{2/3} \varepsilon, \quad z_L = \left(\frac{\pi}{\nu_L}\right)^{2/3} (\nu_L - \varepsilon)$$

1D Harmonic oscillator

$$V(x) = \frac{1}{2}m\omega^2 x^2$$

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n(\xi) e^{-\xi^2/2}, \quad \xi \equiv \sqrt{\frac{m\omega}{\hbar}} x$$

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2, \quad \dots$$

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n = 0, 1, 2, 3, \dots$$

Scattering at Common Potentials

1D δ -function

$$\text{well/barrier} \\ V(x) = \pm\alpha\delta(x)$$

$$R = \frac{1}{1 + \frac{2\hbar^2 E}{m\alpha^2}}, \quad T = \frac{1}{1 + \frac{m\alpha^2}{2\hbar^2 E}}.$$

Hydrogen Atom

The eigenstates $\psi_{nlm}(\vec{r}) = R_{nl}(r)Y_{lm}(\theta, \phi)$

$$R_{nl}(s) = \left[\frac{(n-l-1)!}{2n(n+l)!} \left(\frac{2}{na_0}\right)^3 \right]^{1/2} s^l L_{n-l-1}^{2l+1}(s) e^{-s/2}, \quad s = \frac{2r}{na_0}$$

$$Y_{lm}(\theta, \phi) = (-1)^{m+|m|} \left[\frac{2l+1}{4\pi} \frac{(l-|m|)!}{(l+|m|)!} \right]^{1/2} P_l^{|m|}(\cos\theta) e^{im\phi}$$

$$\text{eigenenergies} \quad E_n = -\frac{\text{Ry}}{n^2} = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$

with associated Legendre and associated Laguerre polynomials:

Associated Laguerre
polynomials

$$L_k^\alpha(z) = \sum_{j=0}^k (-1)^j \binom{k+\alpha}{k-j} \frac{z^j}{j!}$$

Associated Legendre
polynomials

$$P_l^m(z) = \frac{(1-z^2)^{m/2}}{2^l l!} \frac{d^{m+l}}{dz^{m+l}} (z^2 - 1)^l$$

Rydberg energy unit

$$\text{Ry} = 13.6 \text{ eV}$$

Rydberg constant
for hydrogen

$$R_H = 1.0968 \times 10^7 \text{ m}^{-1} \approx 1.1 \times 10^7 \text{ m}^{-1}$$

Rydberg constant
for heavy atoms

$$R_\infty = 1.0974 \times 10^7 \text{ m}^{-1} \approx 1.1 \times 10^7 \text{ m}^{-1}$$